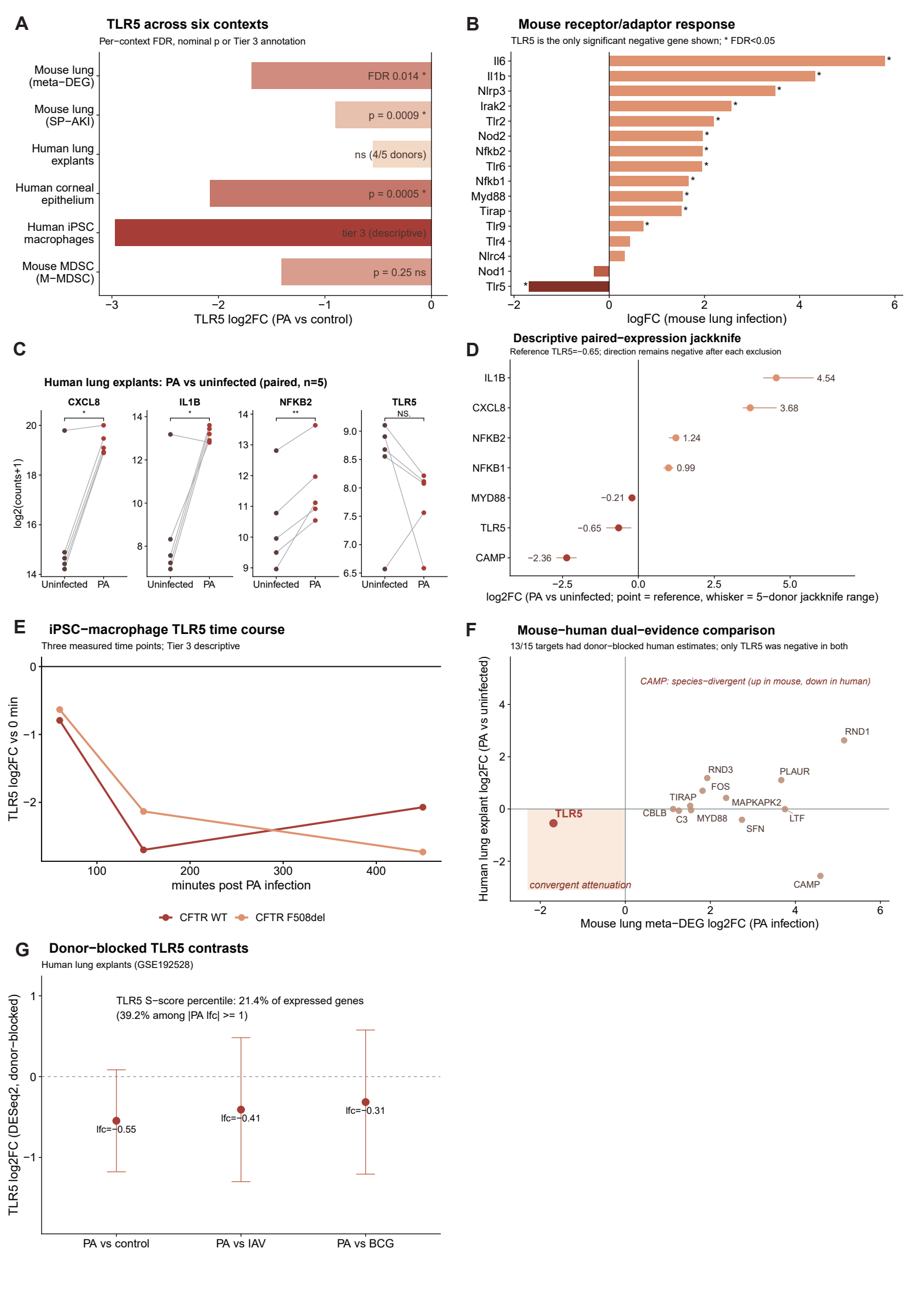

A
TLR5 across six contexts
Per−context FDR, nominal p or Tier 3 annotation
Mouse lung (meta−DEG)
Mouse lung (SP−AKI)
Human lung explants
Human corneal epithelium
Human iPSC macrophages
Mouse MDSC (M−MDSC)
FDR 0.014 *
p = 0.0009 *
ns (4/5 donors)
p = 0.0005 *
tier 3 (descriptive)
p = 0.25 ns
TLR5 log2FC (PA vs control)
B
Mouse receptor/adaptor response
TLR5 is the only significant negative gene shown; * FDR<0.05
Il6
Il1b
Nlrp3
Irak2
Tlr2
Nod2
Nfkb2
Tlr6
Nfkb1
Myd88
Tirap
Tlr9
Tlr4
Nlrc4
Nod1
Tlr5
logFC (mouse lung infection)
C
Human lung explants: PA vs uninfected (paired, n=5)
CXCL8
IL1B
NFKB2
TLR5
NS.
log2(counts+1)
Uninfected
PA
D
Descriptive paired−expression jackknife
Reference TLR5=−0.65; direction remains negative after each exclusion
IL1B
CXCL8
NFKB2
NFKB1
MYD88
TLR5
CAMP
4.54
3.68
1.24
0.99
−0.21
−0.65
−2.36
log2FC (PA vs uninfected; point = reference, whisker = 5−donor jackknife range)
E
iPSC−macrophage TLR5 time course
Three measured time points; Tier 3 descriptive
TLR5 log2FC vs 0 min
minutes post PA infection
CFTR WT
CFTR F508del
F
Mouse−human dual−evidence comparison
13/15 targets had donor−blocked human estimates; only TLR5 was negative in both
CAMP: species−divergent (up in mouse, down in human)
convergent attenuation
RND1
RND3
PLAUR
FOS
TIRAP
MAPKAPK2
CBLB
C3
MYD88
LTF
SFN
TLR5
CAMP
Human lung explant log2FC (PA vs uninfected)
Mouse lung meta−DEG log2FC (PA infection)
G
Donor−blocked TLR5 contrasts
Human lung explants (GSE192528)
TLR5 S−score percentile: 21.4% of expressed genes
(39.2% among |PA lfc| >= 1)
lfc=−0.55
lfc=−0.41
lfc=−0.31
TLR5 log2FC (DESeq2, donor−blocked)
PA vs control
PA vs IAV
PA vs BCG